Auditing a physics-informed augmentation claim: a negative result, and the lever it revealed

Simon Krekels

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Abstract

A thermal solar-module classifier reported a +9.8 percentage-point accuracy gain from heat-equation–based synthetic augmentation. This report audits that claim end to end. Roughly +9 points are a training-schedule artifact; the residual is a class-rebalancing effect that plain image duplication matches, and that does not improve recall on the rare fault classes the method targets. Two diagnostics localise the cause — the synthetic images are trivially separable from real ones (discriminator AUC 0.97), and class weighting already handles the imbalance better than rebalancing. We then *close* the realism gap by perturbing real faults with the heat equation (AUC 0.71), and find the decisive result: even realistic, physically-grounded augmentation gives no rare-class benefit over duplication. Augmentation quality was never the bottleneck. The diagnosis points instead to the training loss: a logit-adjusted long-tail loss raises rare-fault recall from 0.612 to 0.728 (+19% relative, bootstrap CI excluding zero), a tunable trade against majority-class accuracy. The headline was wrong; the investigation that disproved it correctly identified the real lever.

1 Setup and metrics

- **Data.** InfraredSolarModules [1]: ∼20,000 24×40 thermal IR crops, 12 classes, heavily imbalanced (∼50% `No-Anomaly`; four rare fault classes — `Diode-Multi`, `Hot-Spot`, `Hot-Spot-Multi`, `Soiling` — have only ∼120–170 training images each).
- **Model.** EfficientNet-B0 [2] fine-tuned from ImageNet; warmup → cosine schedule; stratified 70/15/15 splits.
- **Harness.** Every comparison is paired on one fixed test set, run over 3 seeds, with bootstrap 95% confidence intervals on *per-class recall*.
- **Key framing.** The objective is rare-*fault detection*, so the figure of merit is **rare-class recall**, not accuracy (which is dominated by the majority class). This distinction drives every conclusion below.

2 The headline is mostly a schedule artifact

The original baseline is undertrained: a slow cosine schedule plus patience-5 early stopping halts it at epoch 12. Fixing only the schedule (a faster 15-epoch cosine) recovers almost the entire gap with *no* synthetic data (Table 1). All later experiments use the 15-epoch schedule.

Run	Schedule	Test acc.
baseline (as reported)	30 ep	0.690
augmented (as reported)	30 ep	0.788
clean, retuned schedule	15 ep	0.784
augmented, retuned schedule	15 ep	0.808

Table 1: The +9.8-point headline is mostly the difference between an undertrained and a well-trained baseline.

3 The backbone choice holds up

EfficientNet-B0 is not assumed: it wins a sweep over four `timm` backbones on identical clean data, *and* is the smallest — the larger backbones overfit this small, imbalanced set (Table 2).

Backbone	Params (M)	Test acc.	Macro F1
EfficientNet-B0	**4.0**	**0.784**	**0.658**
MobileNetV3-Large	4.2	0.760	0.646
ConvNeXt-Tiny	27.8	0.745	0.635
ResNet-50	23.5	0.694	0.577

Table 2: Backbone sweep (clean data, 15-epoch schedule).

4 Honest baselines: physics is no better than oversampling

The decisive comparison is against the *cheap* ways to rebalance. `oversample` duplicates real rare-class images to the same target count as the synthesiser, isolating the synthetic *signal* from rebalancing; `randaugment` is a strong generic-augmentation control. We also toggle class weighting off ("no-weights") to probe its interaction with rebalancing (Table 3).

Condition	Weights	Acc.	Macro F1	Macro rec.	**Rare rec.**
clean	yes	0.782	0.658	0.699	**0.612**
oversample	yes	0.807	0.678	0.697	0.574
randaugment	yes	0.782	0.654	0.702	0.615
physics	yes	0.800	0.668	0.693	0.569
clean	no	0.839	0.699	0.684	0.560
oversample	no	0.826	0.685	0.680	0.582
physics	no	0.837	0.700	0.686	0.568

Table 3: Honest baselines, 15-epoch schedule, 3-seed means. Physics is statistically indistinguishable from `oversample` on every metric, and is the *worst* condition on rare-class recall.

- **Physics does not beat duplication.** `oversample` (0.807 acc) edges out `physics` (0.800); elsewhere they are within noise. The synthetic pixels add nothing over copy-paste.
- **The accuracy gain is a rebalancing artifact.** Physics' +1.9 points over `clean` (3-seed paired $+0.019 \pm 0.006$) come from the *majority* class (`No-Anomaly` recall $0.85 \to 0.88$), not the rare ones. On rare-class recall physics is worst, and significantly below `clean` (paired $\Delta = +0.043$, CI $[0.003, 0.082]$); Fig. 1.
- **Class weighting is the strongest single lever for rare classes.** `clean` with weights (0.612) beats every other condition. Rebalancing is a weaker *substitute* that *hurts* when stacked on weighting.

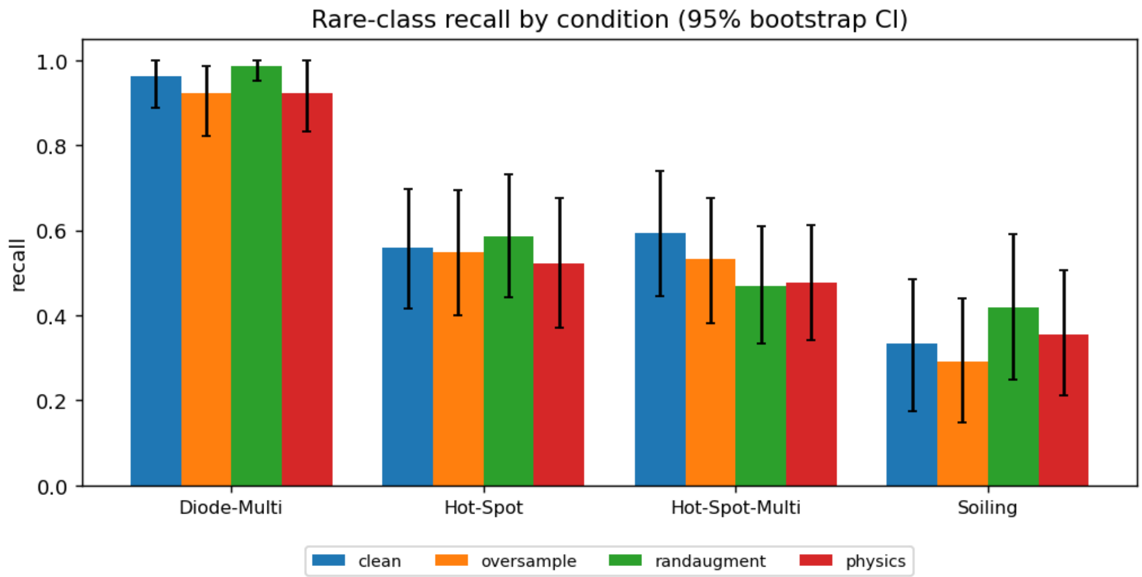

Figure 1: Rare-class recall with 95% bootstrap CIs. Physics (red) is never the best bar. The wide intervals (~ 25 test images per rare class) are themselves a finding: the evaluation is underpowered, which is why a metric-appropriate, CI-based harness was necessary.

5 Why it fails: two diagnostics

D1 — the synthetic images are fake (Fig. 2). A classifier separates synthetic from real images of the same classes at AUC 0.966. The gap *survives heavy blur* (0.934 at radius 2, 0.867 at radius 4), so it lives in the *coarse shape* of the steady-state heat blobs — not seams or compression. The synthetic faults simply do not look like real faults.

D2 — rebalancing is redundant. With class weighting off (Table 3, lower block), accuracy rises and rare recall falls: class weighting and data rebalancing are *substitutes* for handling imbalance, and stacking them over-corrects toward the majority. Even a perfect synthesiser would have to be deployed without fighting the weights.

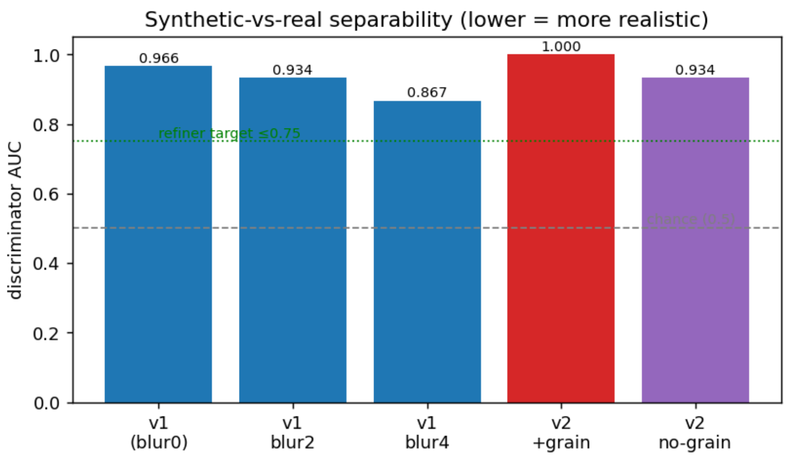

Figure 2: D1 discriminator AUC for the v1 synthesiser under increasing blur (lower = more realistic). The gap persists under blur $\Rightarrow$ it is coarse, structural.

6 Closing the realism gap

If the gap is structural, can we close it? Three attempts, scored by an *independent* discriminator (realism gate, target AUC ≤ 0.75); Fig. 3.

(a) A better hand-built generator (v2) measured worse. v2 replaced steady-state blobs with transient diffusion, fixed an aspect warp, calibrated contrast, and corrected the class models (real `Soiling` is bright and streaky, not a smooth darkening). It looks markedly more realistic (Fig. 4) yet scored AUC 1.000 with naïvely-added i.i.d. grain and 0.934 without — visual realism was misleading, and the additive-blend-on-clean-base paradigm is near a detectability ceiling.

(b) A learned refiner (SimGAN) failed. A residual refiner trained adversarially with a self-regularization term [3] either collapsed to near-identity (patch discriminator) or added artifacts that pushed AUC to 0.980 (global discriminator). The reason is principled: SimGAN refines *appearance* assuming the structure is right, but D1 showed the gap is *structural* — which a self-regularized refiner cannot change.

(c) Perturbing real faults passed. The heat equation's time-evolution operator is Gaussian diffusion, so we start from a *real* fault crop, decompose it into background + fault + grain, and physically perturb only the fault (diffuse = thermal time, rescale = severity), keeping the real background and grain (Fig. 5). Structure, texture and grain are real by construction; AUC drops to **0.711**, clearing the gate — the first generator to do so.

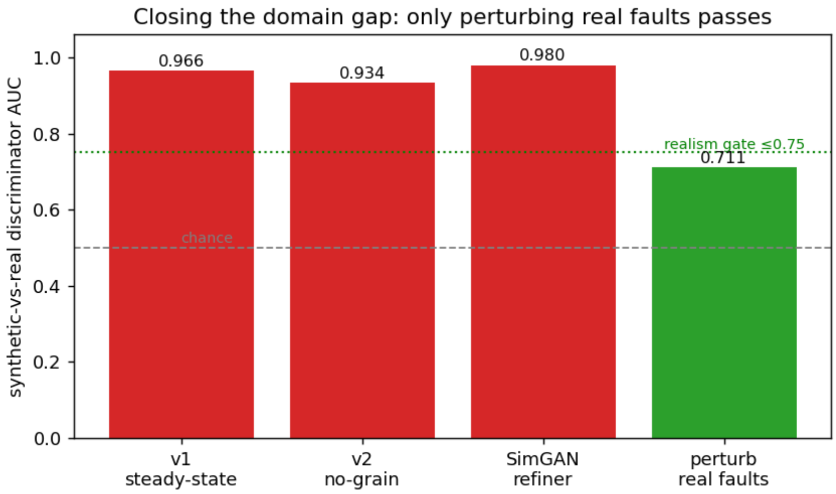

Figure 3: Realism gate across approaches. Only perturbing real faults closes the domain gap.

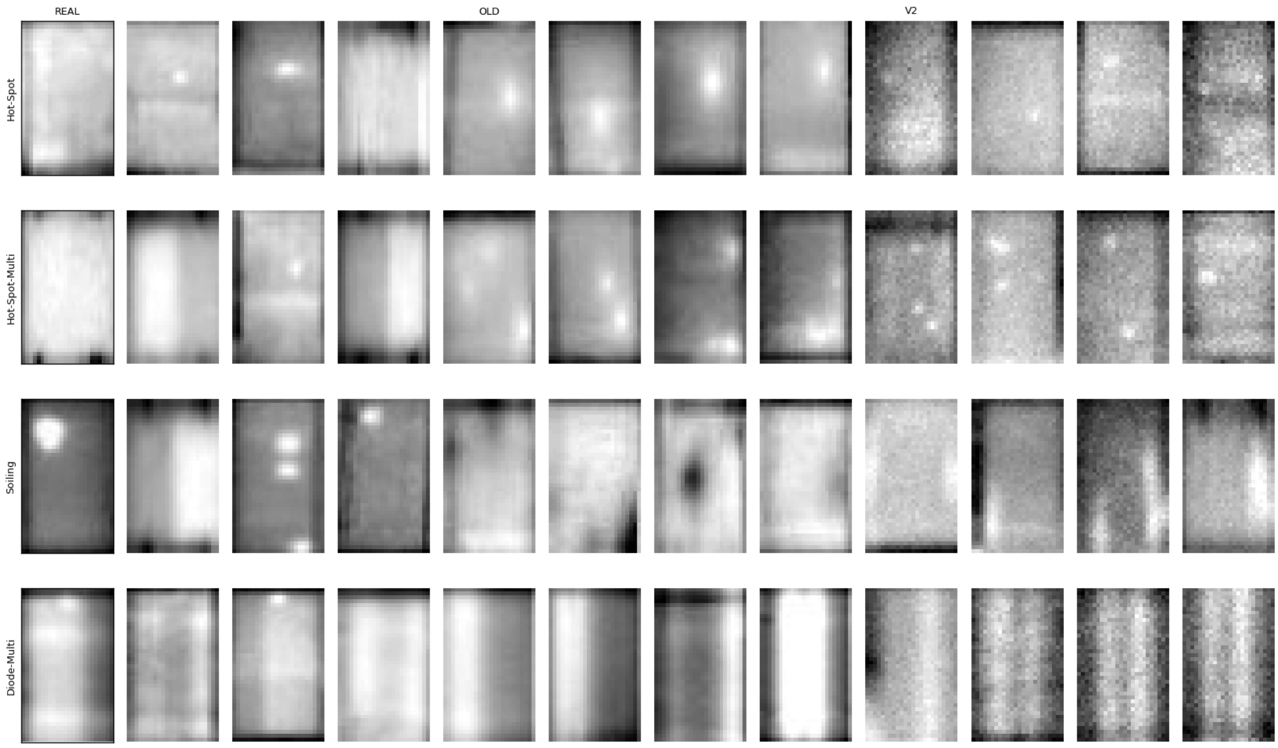

Figure 4: Real | v1 | v2 synthetic. v2 is visually closer but measured more separable — a cautionary result.

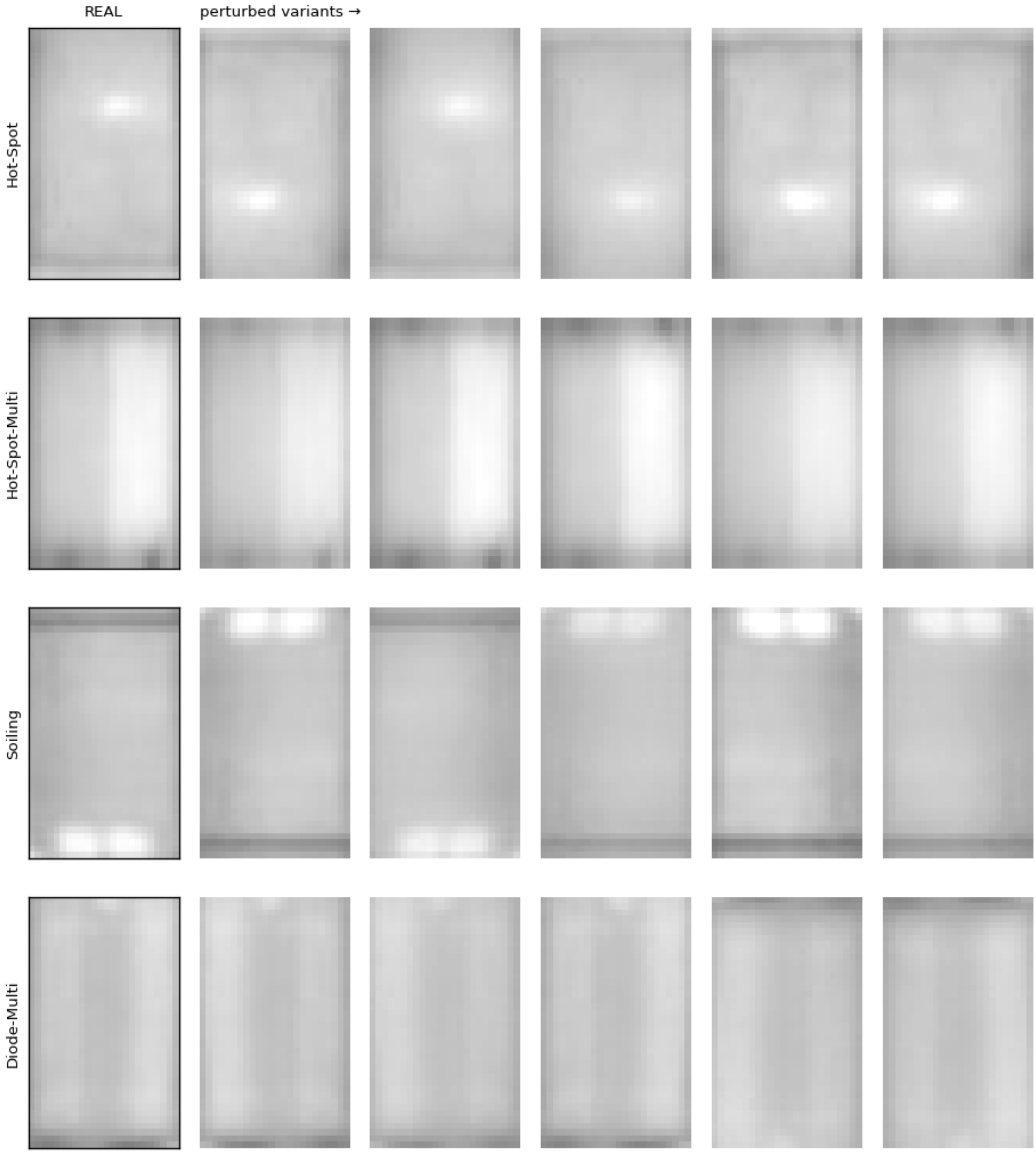

Figure 5: A real fault (left) and physics-perturbed variants. Real background, frame and grain are preserved; diversity is in the physical axes (severity, spread, orientation).

7 The decisive result: realism is not utility

With the realism gap finally closed, does it help? No. On rare-class recall, `perturb` (0.572) is statistically identical to `oversample` (0.574): paired $\Delta = -0.0015$, CI $[-0.038, +0.035]$, in both weight regimes, and still below `clean` (0.612).

This is the central finding. We made augmentation *realistic* and it *still* gave no benefit over duplication — so augmentation quality was never the bottleneck. The binding constraints are (i) $\sim$150 real images per rare class is too little information for any augmentation to manufacture new discriminative signal, and (ii) class weighting already handles the imbalance. For rare-fault detection on this dataset, augmentation — synthetic or real-derived, crude or realistic — is *not the lever*.

8 The lever: a long-tail loss

The diagnosis points at the training loss, not the data. Logit adjustment [4] adds $\tau \log(\text{prior})$ to the logits during training and infers on raw logits, shifting the decision boundary by the class priors rather than reweighting gradients. The strength τ is a knob on the recall/accuracy frontier (Fig. 6, Table 4).

Loss	Acc.	Macro rec.	**Rare rec.**	Δ rare vs clean (CI)
clean (weighted CE)	0.782	0.699	0.612	—
logit-adj $\tau = 1.0$	**0.808**	0.709	0.634	$+0.022$ $[-0.016, +0.058]$
logit-adj $\tau = 1.5$	0.763	**0.716**	0.696	**$+0.084$ $[+0.038, +0.131]$**
logit-adj $\tau = 2.0$	0.683	0.703	**0.728**	**$+0.116$ $[+0.071, +0.162]$**

Table 4: Logit adjustment, 15-epoch schedule, 3-seed means. Rare-fault recall rises 0.612 → 0.728; for $\tau \geq 1.5$ the gain is statistically significant (CI excludes zero), unlike every augmentation condition.

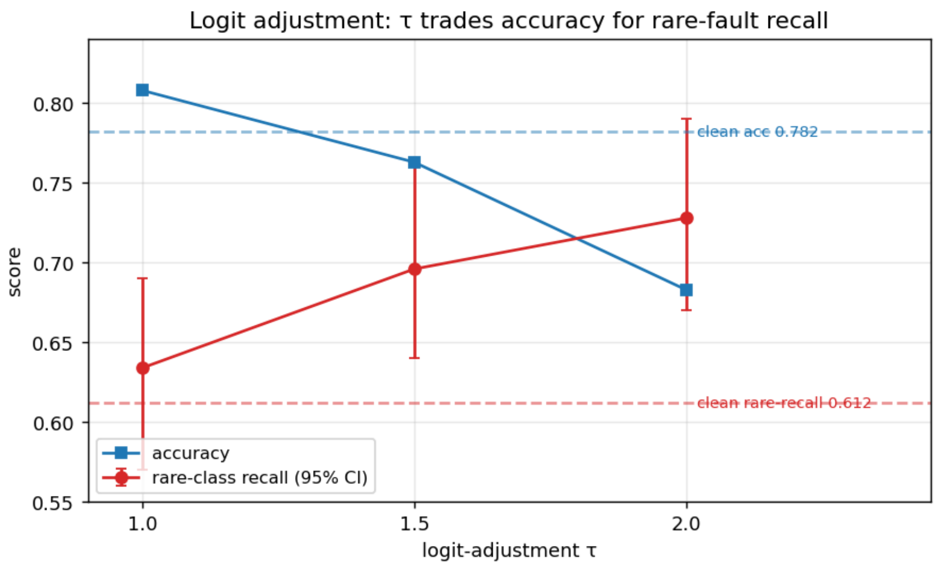

Figure 6: τ trades majority-class accuracy for rare-fault recall along a clean frontier. $\tau = 1.5$ is a balanced default (best macro-recall of any condition, 0.716); $\tau = 2.0$ maximises detection.

This is the first *statistically significant* improvement on the objective metric in the whole study. For a fault-detection task, where a missed fault costs more than a false alarm, the trade is favourable: $\tau = 1.5$ raises rare recall to 0.696 at a small accuracy cost (0.763 vs 0.782), and $\tau = 2.0$ to 0.728 if false alarms are acceptable.

9 Discussion and recommendations

- **The right role for physics is structure, not pixels.** Hand-built PDE generators failed because they were asked to render appearance, which is hard to specify; perturbing real faults succeeded at realism because it kept real appearance and used physics only for the fault's transformation.

- **But here, even that was not enough** — the limiting factor was information content and the loss, not generation. This is specific to the regime (scarce rare classes; effective class weighting), not a universal claim about augmentation.
- **Recommendation.** Adopt logit adjustment ($\tau \approx 1.5$) as the default loss, replacing inverse-frequency weighting; expose τ as the recall/precision operating-point control. Further headroom: LDAM with deferred reweighting [5], focal loss [6] (with care — it can over-focus on the ambiguous `Soiling` class), on-domain self-supervised pretraining, and collecting more real rare-fault labels.
- **Methodological takeaway.** The honest-baseline plus bootstrap-CI harness turned a fragile +9.8 headline into a bounded negative result *and* pointed to the real lever. A well-characterised negative result, with the limiting factor identified, was more valuable than the inflated positive it replaced.

References

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[4] A. K. Menon, S. Jayasumana, A. S. Rawat, H. Jain, A. Veit, and S. Kumar, "Long-tail learning via logit adjustment", in International conference on learning representations (iclr) (2021), arXiv:2007.07314.

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